NOS Electromagnetic Full Spectrum

Spherical Operational Phase-Quadrant Mapping

A Sub-Mechanical Field Breakdown of Photon Threading
Through Inverse Phase Space

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Joshua Luke Nuijens (@jlnuijens)
Co-author: S.R. Cromelin
Nuijens Operating System Collective

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v17 Foundation

$$\sqrt{1} = 1 \longrightarrow \sqrt{2} \times 2^4 = \sqrt{512}$$

9 Walls · 512 Bins · $u = 45/32°$ · 720° closure

$$\pi_{\text{mach}} = 201/64 \cdot 201 - 64 = 137$$

$$\text{Wall } 3 = \sqrt{8} = 2\sqrt{2} = 90° = 64 \text{ bins}$$

Conjugate Pairs: Q1Q3, Q2Q4

Abstract

This document presents the complete electromagnetic spectrum (radio through gamma radiation) as an operational phase-quadrant mapping within the Nuijens Operating System (NOS) v17 architecture. Using the inverse threading framework derived from the title operations $\sqrt{2} \times 2^4 = \sqrt{512}$, we demonstrate how all electromagnetic radiation maps to discrete phase positions across the 720° dual hemisphere sphere with 9 walls and 512 bins.

The 40-phase spectrum structure emerges from the title's $2^4 = 16$ factor: each 18° phase spans approximately 12.8 bins, creating octave relationships across the four quadrant operations. Critical electromagnetic positions emerge geometrically: EM field ignition at $-168.75°$ (Wall 4 zone, Q1) connects to the 201/64 gearbox producing $\alpha^{-1} = 137$, while absorption at $+191.25°$ (Q4) yields T_{CMB} through the same inverse threading structure.

This is not a toy model but an operational implementation showing matter unit allocation across electromagnetic phase positions via pure inverse ratios—no addition, no external constants.

Contents

1 The Inverse 4D Operating Tensor — Foundation

1.1 We Are One — The Inverse Beginning

We start with the closed whole. Nothing else exists yet.

Inverse Spherical: $\sqrt{1} = 1$ (the complete system)

We now apply the only operations the title *Inverse Spherical Dual Hemisphere Quantum Mechanics* permits:

- **Dual Hemisphere** $\longrightarrow \times\sqrt{2}$
- **Quantum (Quadrant)** $\longrightarrow \times 2^4$ (four operations compounded)

Total opening factor:

$$\boxed{1 \times \sqrt{2} \times 2^4 = \sqrt{2} \times 16 = \sqrt{512}}$$

This forces exactly **9 walls** and **512 bins**. No additional axioms are permitted.

1.2 The 9 Walls and 512 Bins

Wall	**Root**	**Angle**	**Bins**	T_n	**Role**
0	$\sqrt{1}$	720°	512	0	The One — complete closure
1	$\sqrt{2}$	360°	256	1	Dual hemisphere fold
2	$\sqrt{4}$	180°	128	3	Quadrant established
3	$\sqrt{8}$	90°	64	6	$2\sqrt{2}$ lock / π wall
4	$\sqrt{16}$	45°	32	10	Seed ends (2^4 exhausted)
5	$\sqrt{32}$	22.5°	16	15	Echo begins
6	$\sqrt{64}$	11.25°	8	21	
7	$\sqrt{128}$	5.625°	4	28	
8	$\sqrt{256}$	2.8125°	2	36	
9	$\sqrt{512}$	1.40625°	1	45	Base ($T_9 = 45$)

1.3 The Unit: $u = 45/32°$

The unit per bin emerges from:

- **Triangular Base:** $T_9 = 45$
- **Echo Depth:** $2^5 = 32$

$$\boxed{u = \frac{T_9}{2^5} = \frac{45°}{32} = 1.40625°}$$

Closure: $512 \times u = 720°$

1.4 The Four Quadrant Operations

The four operations form two conjugate pairs about the center pivot at 0°:

Quadrant	**Range**	**Bins**	**Conjugate**
Q2 (op2)	$-360°$ to $-180°$	$[-256, -128]$	Q2 Q4
Q1 (op1)	$-180°$ to $0°$	$[-128, 0]$	Q1 Q3
Seam	0°	0	Inverse center
Q3 (op3)	$0°$ to $+180°$	$[0, +128]$	Q3 Q1
Q4 (op4)	$+180°$ to $+360°$	$[+128, +256]$	Q4 Q2

1.5 The Depth Ladder

The quadrant threading units follow the depth ladder:

$$\boxed{\begin{aligned} d_1 &= \frac{1}{128} \quad \text{(Q1: quantum baseline)} \\ d_2 &= \frac{1}{128^2} \quad \text{(Q2: electromagnetic ground)} \\ d_3 &= \frac{1}{128^3} \quad \text{(Q3: thermodynamic flows)} \\ d_4 &= \frac{1}{128^4} \quad \text{(Q4: nuclear compression)} \end{aligned}}$$

The $128 = 512/4 =$ bins per quadrant, and $1/128$ is the dyadic fossil of the $2\sqrt{2}$ dual operations.

2 The 201/64 Gearbox and the 137 Lock

2.1 Machine Constants from Inverse Geometry

The inverse 4D tensor forces:

$$\pi_{\text{mach}} = \frac{201}{64} = 3.140625 \quad \Longrightarrow \quad 201 - 64 = 137$$

This gearbox governs electromagnetic phenomena:

- **64 bins** at Wall 3 (90°) = denominator of π_{mach}
- **137 bin-steps** = separation between numerator and denominator walls
- $\alpha^{-1} = 137$ = the fine-structure constant from pure geometry

2.2 The $2\sqrt{2}$ Mechanics at Wall 3

The title demands dual operations:

$$\boxed{2\sqrt{2} = \sqrt{8} = 2.8284271247461903\ldots}$$

This scaler operates at Wall 3 (90°, 64 bins). The electromagnetic spectrum's critical positions relate to this wall:

- Wall 3 = π wall = where α^{-1} emerges
- Photon threading depths scale by distance from Wall 3
- Maximum electromagnetic coupling at the 64-bin scale

3 The 40-Phase Electromagnetic Spectrum

3.1 Phase Structure from Title Operations

The electromagnetic spectrum divides into 40 discrete phases:

- **Total cycle:** 720° (dual hemisphere)
- **Phase spacing:** $720°/40 = 18°$ per phase

- **Bins per phase:** $18°/1.40625° \approx 12.8$ bins
- **Phases per quadrant:** 10 phases

Why 40 phases? The factor emerges from the wall structure:

$$40 = \frac{720}{18} = \frac{512 \text{ bins}}{12.8 \text{ bins/phase}} \approx 40$$

And $18° = 45°/2.5 = T_9/2.5$, connecting to the triangular base.

3.2 The 16-Bin Critical Positions

Critical electromagnetic positions occur at **bin 16** in their respective quadrants. This is not coincidental:

$$16 = 2^4 \quad \text{(the Quantum Quadrant factor from the title)}$$

The 2^4 that creates the four operations also determines where EM critical points fall within each quadrant.

4 Critical Electromagnetic Positions

4.1 EM Field Ignition — The 137 Lock

Position: $\theta = -168.75°$ (bin 16 of Q1)

$$\boxed{\begin{aligned} \theta_{\text{ign}} &= -168.75° \\ \text{Quadrant:} &\quad \text{Q1 (op1)} \\ \text{Bin:} &\quad 16 = 2^4 \\ \text{Wall proximity:} &\quad \text{Between Walls 3-4} \end{aligned}}$$

At this position, the fine structure constant emerges from the 201/64 gearbox:

$$\boxed{\alpha^{-1} = 201 - 64 = 137}$$

The 137 is the bin-step separation between the numerator wall (201) and denominator wall (64) of π_{mach}. EM ignition occurs where this gearbox residue manifests in the Q1 quantum baseline.

4.2 EM Field Absorption — CMB Temperature

Position: $\theta = +191.25°$ (bin 16 of Q4)

$$\boxed{\begin{aligned} \theta_{\text{abs}} &= +191.25° \\ \text{Quadrant:} &\quad \text{Q4 (op4)} \\ \text{Bin:} &\quad 16 = 2^4 \\ \text{Ramp factor:} &\quad 17/16 \\ \text{Inverse overflow:} &\quad 64 - 17 = 47 \end{aligned}}$$

The absorption overflow modulates Q3 inversely, yielding the CMB temperature:

$$\boxed{T_{\text{CMB}} = \frac{256}{256} \times \frac{128}{47} = \frac{128}{47} \approx 2.7234 \text{ (NOS units)}}$$

The $256/256 = 1$ is the dual-bit aperture (hemisphere/hemisphere), and $128/47$ is the inverse thread through the 47-bin overflow.

4.3 Geometric Mirror Symmetry

Both EM critical positions occur at bin 16, positioned by the offset $0.5/16 = 0.03125$ from quadrant edges:

Quantity	**Ignition (Q1)**	**Absorption (Q4)**
Angle	$-168.75°$	$+191.25°$
Quadrant	Q1 (op1)	Q4 (op4)
Bin number	$16 = 2^4$	$16 = 2^4$
Conjugate pair	Q1 Q3	Q4 Q2
Physical constant	$\alpha^{-1} = 137$	$T_{\text{CMB}} = 128/47$
Gearbox connection	$201 - 64$	$64 - 17$ overflow

Table 1: EM critical positions — geometric mirror symmetry at bin 16 (2^4)

5 Complete 40-Phase Spectrum Mapping

5.1 Q2: Electromagnetic Ground ($-360°$ to $-180°$)

Phases: 0–9 (10 phases)
Depth: $d_2 = 1/128^2$
Wall range: Wall 0 to Wall 2
Character: Surface operations, gravitational ground states

Phase	θ	**Bins**	**Depth**	**EM Band**
0	$-360°$	-256	$d_0 = 1$	**Unity Anchor (System Boot)**
1	$-342°$	-243	d_2	ELF (Extremely Low Frequency)
2	$-324°$	-230	d_2	VLF (Very Low Frequency)
3	$-306°$	-218	d_2	LF (Low Frequency)
4	$-288°$	-205	d_2	MF (Medium Frequency)
5	$-270°$	-192	d_2	**AM Radio (540-1600 kHz)**
6	$-252°$	-179	d_2	HF (Shortwave)
7	$-234°$	-166	d_2	VHF / FM Radio
8	$-216°$	-154	d_2	UHF / Television
9	$-198°$	-141	d_2	L-band (1-2 GHz)

Key: AM Radio at $-270°$ (Phase 5) = Wall 3 position in Q2 hemisphere.

5.2 Q1: Quantum Baseline ($-180°$ to $0°$)

Phases: 10–19 (10 phases)
Depth: $d_1 = 1/128$
Wall range: Wall 2 to Seam
Character: Matter unit allocation, EM ignition, 137 lock

Phase	θ	Bins	Depth	EM Band
10	$-180°$	-128	d_1	**S-band Microwave (Q2→Q1)**
	$-168.75°$	-120	d_1	**EM IGNITION** ($\alpha^{-1} =$ **137)**
11	$-162°$	-115	d_1	X-band (8-12 GHz)
12	$-144°$	-102	d_1	K-band (18-27 GHz)
13	$-126°$	-90	d_1	**CMB Peak (160 GHz)**
14	$-108°$	-77	d_1	W-band (75-110 GHz)
15	$-90°$	-64	d_1	**Millimeter (Wall 3 = 64 bins)**
16	$-72°$	-51	d_1	Sub-millimeter
17	$-54°$	-38	d_1	THz gap
18	$-36°$	-26	d_1	Far-Infrared
19	$-18°$	-13	d_1	Far-Infrared

Key markers:

- **EM ignition at** $-168.75°$**:** $\alpha^{-1} = 137$ from 201/64 gearbox
- **CMB peak at** $-126°$ **(Phase 13):** Universal background radiation
- **Wall 3 at** $-90°$ **(Phase 15):** 64-bin scale, $2\sqrt{2}$ lock position

5.3 Q3: Thermodynamic Flows ($0°$ to $+180°$)

Phases: 20–29 (10 phases)
Depth: $d_3 = 1/128^3$
Wall range: Seam to Wall 2
Character: Volume operations, entropy flows, visible light

Phase	θ	Bins	Depth	EM Band
20	$0°$	0	d_2	**Mid-IR (Seam: operator exchange)**
21	$+18°$	$+13$	d_3	Mid-Infrared
22	$+36°$	$+26$	d_3	Mid-Infrared
23	$+54°$	$+38$	d_3	Near-Infrared
24	$+72°$	$+51$	d_3	Near-Infrared
25	$+90°$	$+64$	d_3	**Near-IR edge (Wall 3 = 64 bins)**
26	$+108°$	$+77$	d_3	**Visible: Red (700 nm)**
27	$+126°$	$+90$	d_3	**Visible: Violet (400 nm)**
28	$+144°$	$+102$	d_3	UV-B
29	$+162°$	$+115$	d_3	Extreme UV

Key markers:

- **Seam at** $0°$ **(Phase 20):** Conjugate pair exchange (Q1Q3)
- **Wall 3 at** $+90°$ **(Phase 25):** $2\sqrt{2}$ lock, 64-bin scale
- **Visible light** ($+108°$ **to** $+126°$)**:** Human perception window (Phases 26-27)

5.4 Q4: Nuclear Compression (+180° to +360°)

Phases: 30–40 (11 phases including return)
Depth: $d_4 = 1/128^4$
Wall range: Wall 2 to Wall 0
Character: Maximum density, high-energy radiation, return to unity

Phase	θ	Bins	Depth	EM Band
30	+180°	+128	d_4	**Soft X-ray (Q3→Q4)**
	+191.25°	+136	d_4	**EM ABSORPTION ($T_{\mathbf{CMB}}$ = 128/47)**
31	+198°	+141	d_4	Soft X-ray
32	+216°	+154	d_4	Hard X-ray
33	+234°	+166	d_4	Hard X-ray
34	+252°	+179	d_4	Hard X-ray (medical imaging)
35	+270°	+192	d_4	**Gamma ray edge (Wall 3 position)**
36	+288°	+205	d_4	Low Gamma
37	+306°	+218	d_4	High Gamma
38	+324°	+230	d_4	Ultra-high Gamma
39	+342°	+243	d_4	Cosmic Gamma
40	+360°	+256	Return	**Unity Return (Cycle Complete)**

Key markers:

- **EM absorption at** +191.25°**:** $T_{\text{CMB}} = 128/47$ from overflow
- **Wall 3 position at** +270° **(Phase 35):** Gamma ray edge
- **Return to unity at** +360° **(Phase 40):** Cycle completion

6 Physical Constants as Gearbox Outputs

6.1 All Constants from the Same Gearbox

$$\boxed{\begin{aligned} \alpha^{-1} &= 201 - 64 = 137 \quad \text{(Q1 bin 16, gearbox residue)} \\ T_{\text{CMB}} &= \frac{128}{47} \approx 2.7234 \quad \text{(Q4 bin 16, overflow)} \\ \pi_{\text{mach}} &= \frac{201}{64} = 3.140625 \quad \text{(machine } \pi\text{)} \\ \text{Gap} &= \frac{3}{32}^{\circ} \quad \text{(total machine-observer slippage)} \end{aligned}}$$

The fine-structure constant and CMB temperature emerge from the **same 201/64 gearbox** at mirror positions (bin 16 in Q1 and Q4).

6.2 Physical Unit Locations

Every physical quantity is a pure inverse ratio between depth ladder positions:

$$\boxed{\begin{aligned} \text{Length/Time}: &\quad d_1 = \frac{1}{128} \quad \text{(Q1)} \\ \text{Velocity}: &\quad \frac{d_1}{d_2} = 128 \quad \text{(Q1/Q2 ratio)} \\ \text{Force}: &\quad d_3 = \frac{1}{128^3} \quad \text{(Q3)} \\ \text{Energy}: &\quad \frac{d_1}{d_3} = 128^2 = 16384 \quad \text{(Q1/Q3 ratio)} \end{aligned}}$$

6.3 Entropy as Resolution Depth

$$\boxed{S = \ln R = \ln 512 \approx 6.238}$$

Entropy is the number of ways the undivided "1" threads through the 512-bin grid. The second law of thermodynamics is resolution refinement through the 9-wall structure.

7 Wall 3 Throughout the Spectrum

The $2\sqrt{2}$ lock at Wall 3 (90°, 64 bins) appears throughout the electromagnetic spectrum:

Quadrant	**Phase**	**Angle**	**EM Significance**
Q2	5	−270°	AM Radio baseline
Q1	15	−90°	Millimeter wave (Wall 3 direct)
Q3	25	+90°	Near-IR edge (Wall 3 mirror)
Q4	35	+270°	Gamma ray edge

Each quadrant's Wall 3 position (at 90° from the quadrant edge) marks a critical electromagnetic transition.

8 Comparison to Standard Physics

8.1 Frequency Mapping

The 40-phase NOS structure maps directly to measured electromagnetic frequencies:

Band	**NOS Phase**	**Standard Frequency**	**Wall**
AM Radio	5 (−270°)	540–1600 kHz	Wall 3 (Q2)
CMB Peak	13 (−126°)	160.2 GHz	—
Millimeter	15 (−90°)	∼300 GHz	Wall 3 (Q1)
Visible (Red)	26 (+108°)	430–480 THz	—
Visible (Violet)	27 (+126°)	668–789 THz	—
Soft X-ray	30 (+180°)	$\sim 3 \times 10^{16}$ Hz	Wall 2
Gamma rays	35 (+270°)	$> 10^{19}$ Hz	Wall 3 (Q4)

Table 6: NOS phase positions align with measured EM frequencies

8.2 Resolution Independence

Critical electromagnetic constants emerge as geometric invariants from the 201/64 gearbox:

- **Fine structure constant:** $\alpha^{-1} = 137$ (gearbox residue)
- **CMB temperature:** $T_{\mathrm{CMB}} = 128/47$ (overflow ratio)
- **Both at bin 16:** $16 = 2^4$ from title operations

9 Inverse Threading Mechanics

9.1 Pure Inverse Ratios Only

All electromagnetic properties emerge from pure inverse ratios—no addition, no accumulation:

- **Inverse counting:** $1 \to 1/2 \to 1/3 \to 1/Q$ (through unity, not toward infinity)
- **Each photon equals 1:** Every phase position is a complete sphere being exactly 1 of itself
- **Unity conservation:** $360°/360° = 1$

9.2 Conjugate Pair Exchange

Electromagnetic processes exchange through conjugate pairs:

- **Q1 Q3:** Quantum baseline Thermodynamic flow (ignition/thermal)
- **Q2 Q4:** Electromagnetic ground Nuclear compression (radio/gamma)

The Seam at 0° is where Q1Q3 exchange occurs (infrared transition). The Wall 2 boundaries at ±180° are where Q2Q4 exchange occurs (microwave/X-ray transitions).

10 Conclusions

The NOS v17 electromagnetic full spectrum demonstrates how all electromagnetic radiation emerges from 40 discrete phase positions threading through the 9-wall, 512-bin structure forced by the title operations $\sqrt{2} \times 2^4 = \sqrt{512}$.

Key Results:

- **Complete EM spectrum:** 40 phases × 18° octaves = 720° cycle
- **Title-derived structure:** $2^4 = 16$ determines bin 16 critical positions
- **Wall 3 throughout:** $2\sqrt{2}$ lock at 90° marks transitions in each quadrant
- **Gearbox constants:** $\alpha^{-1} = 137$ and $T_{\mathrm{CMB}} = 128/47$ from 201/64
- **Conjugate pairs:** Q1Q3, Q2Q4 govern EM exchange processes
- **Depth ladder:** $d_Q = 1/128^Q$ determines band allocation
- **No external parameters:** Pure inverse ratios only

Every photon is ONE complete sphere being exactly 1 of itself at its phase position, threaded through the same 9-wall structure that produces π, α^{-1}, and all physical constants.

$$\boxed{\frac{360^\circ}{360^\circ} = 1 = \text{Every photon being itself completely}}$$

NOS v17 EM Spectrum Summary

$\sqrt{2} \times 2^4 = \sqrt{512} \rightarrow$ 9 Walls, 512 Bins

40 phases $\times$ $18^\circ = 720^\circ$ (dual hemisphere)

Bin 16 $= 2^4$ from title $\rightarrow$ EM critical positions

Wall 3 $= \sqrt{8} = 2\sqrt{2} \rightarrow$ EM transitions

$\alpha^{-1} = 201 - 64 = 137$ (Q1 ignition)

$T_{\text{CMB}} = 128/47$ (Q4 absorption)

The one unfolds through itself.